A Study of International Standards for Safe and Responsible AI Governance in Organizations

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ABSTRACT

The accelerating advancement of Artificial Intelligence (AI), particularly large language models (LLMs), has generated significant concerns regarding the safety and responsibility of its use. Governments and organizations, such as the Brazilian Judiciary, have responded with specific regulations. This article examines the safe and responsible AI controls established by National Council of Justice (CNJ) Resolution No. 615/2025 and cross-references them with international risk management and management systems frameworks, notably International Organization For Standardization And International Electrotechnical Commission (ISO/IEC) 42001:2023 and the National Institute Of Standards And Technology (NIST) Artificial Intelligence Risk Management Framework (AI RMF) 1.0. It demonstrates how the normative guidelines of the Judiciary can be operationalized and audited through the adoption and adaptation of global practices, focusing on critical aspects such as algorithmic impact assessment, data governance, transparency, and bias mitigation.

CCS CONCEPTS

• Computing methodologies • Artificial intelligence • Control methods • Computational control theory

KEYWORDS

Artificial Intelligence, CNJ, ISO, NIST

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1. Introduction

In Brazil, the Judiciary, recognizing the rapid development of AI, especially generative models, established National Council of Justice (CNJ) Resolution No. 615, of March 11, 2025, to regulate the development, use, and governance of AI solutions. Resolution 615/2025 seeks to promote technological innovation in a safe, transparent, non-discriminatory, and ethical manner, establishing the need for auditing AI solutions from the perspectives of information security, data protection, performance, robustness, reliability, and the prevention of discriminatory biases [1]. Operationalizing these legal and ethical requirements is a central challenge for Responsible AI Engineering.

The article aims to establish a bridge between national regulatory mandates (CNJ 615/2025) and international frameworks that provide concrete guides for AI risk management and the implementation of AI Management Systems (AIMS), such as International Organization For Standardization And International Electrotechnical Commission (ISO/IEC) 42001:2023 and the National Institute Of Standards And Technology (NIST) Artificial Intelligence Risk Management Framework (AI RMF) 1.0.

1.1. Problem Statement

The proliferation of principles-based AI regulations has created a significant operationalization challenge for organizations. Normative mandates, such as those articulated in CNJ Resolution 615/2025, define high-level ethical and legal requirements (e.g., fairness, transparency, human oversight) but often lack the prescriptive detail necessary for direct technical implementation and auditing. This creates a critical normative-prescriptive gap, where legal obligations are established without a clear, standardized methodology for their fulfillment in the AI lifecycle.

The core problem, therefore, is the absence of a structured and verifiable pathway to translate the abstract legal requirements of the Brazilian Judiciary into concrete organizational and technical controls. This deficiency poses a substantial risk to effective governance, as it impedes the ability of organizations to systematically implement, measure, and demonstrate compliance, thereby undermining the foundational objective of ensuring safe and responsible AI systems.

1.2. Research Objectives

To solve the problem identified above, this paper sets out to build a practical bridge between the Brazilian regulation and proven international standards. Our goal is to provide a clear roadmap for anyone tasked with building or overseeing AI in this context.

Specifically, this study aims to: break down the legal language of CNJ Resolution 615/2025 into clear, actionable requirements; explore the ISO/IEC 42001 and NIST AI RMF 1.0 frameworks to find the specific tools, processes, and controls they offer for AI

governance; connect the dots by systematically mapping the requirements from the CNJ regulation to the practical solutions found in the international standards, showing exactly how compliance can be achieved and audited.

2. Brazilian Regulatory Context: CNJ Resolution 615/2025

The fundamentals and principles of Resolution 615/2025 highlight the central role of ethics and responsibility:

- Human Centrality and Supervision: The use of AI in the Judiciary must respect the centrality of the human person and requires effective, periodic, and adequate human supervision at all stages of the life cycle. It is crucial that the use of AI be auxiliary and complementary, and its use as an autonomous instrument for judicial decision-making is prohibited. The magistrate remains fully responsible for the decisions.
- Transparency and Explainability: Transparency will be promoted through clear indicators and public reports, using simple language. AI must ensure auditability (the ability to evaluate algorithms, data, and results), explainability (understanding how "decisions" are made), and contestability (the possibility of questioning and reviewing results).
- Justice and Non-Discrimination: Promoting equality, plurality, and decisional justice is fundamental, with preventive measures against the emergence of illegal or abusive discriminatory biases. If a discriminatory bias is verified and cannot be corrected, the AI solution must be discontinued.

Governance is exercised by the National Committee for Artificial Intelligence of the Judiciary, whose competencies include assessing the need to update risk categorization, establishing governance rules for the Sinapses system, and determining the performance of audits and monitoring.

2.2 Risk Classification and Management

CNJ Resolution 615/2025 requires courts to conduct a preliminary assessment of AI solutions to define their risk level, categorizing them as high or low risk. This categorization is based on the potential impact on fundamental rights, model complexity, and the quantity of sensitive data used.

- High-Risk (HR) solutions (e.g., identification of behavioral profiles and patterns—AR1, valuation of evidence—AR2, or formulation of conclusive judgments on the application of legal norms—AR4) must be subjected to regular audit processes and continuous monitoring. For these, the performance of the Algorithmic Impact Assessment (AIA) is mandatory—a continuous process that analyzes impacts on fundamental rights, identifying preventive and mitigating damage measures.
- Low-Risk (LR) solutions (e.g., execution of ordinary procedural acts, data classification and grouping—BR1, detection of decisional patterns—BR2, or production of supporting texts—BR4) must be periodically monitored and reviewed.

There are also explicit prohibitions that characterize excessive risk, such as solutions that: do not allow human review of proposed results; evaluate personality traits or behaviors to predict crimes; or classify natural persons to assess the plausibility of their rights. These solutions must be discontinued.

2.3 Data and Information Security Controls

Personal data protection and information security are pillars of Resolution 615/2025.

- Training data must be representative and subject to quality curation. The adoption of secure, traceable, and auditable data sources is required.
- Courts must implement mechanisms for "privacy by design" (privacy preservation from conception) and "privacy by default" (using a high level of confidentiality by default).
- Data anonymization is mandatory for confidential data or data protected by judicial secrecy.
- The use of confidential data or data protected by judicial secrecy to process, analyze, or generate content in private/external LLMs or generative AI systems is prohibited.
- The storage and execution of AI must occur in environments that guarantee data isolation and meet consolidated information security standards, with reference to international standards like ISO/IEC 42001 and those of NIST being recommended.

3. International Standards for AI Risk Management

The analysis and cross-referencing of CNJ guidelines with ISO/IEC 42001 and the NIST AI RMF 1.0 provide the structure for implementing formal controls.

3.1 ISO/IEC 42001: The AI Management System

ISO/IEC 42001:2023 is an International Standard that specifies requirements and provides guidance for establishing, implementing, maintaining, and continually improving the AIMS [2]. The AIMS follows the High-Level Structure (HLS) of ISO standards (such as ISO 27001, mentioned as a system that can be integrated), encompassing:

- Context of the Organization (Clause 4): Requires understanding relevant external and internal issues, as well as the needs and expectations of interested parties (AI users, suppliers, regulators, affected individuals).
- Leadership (Clause 5): Top management must demonstrate leadership and commitment, establishing the AI Policy.
- Planning (Clause 6): Includes actions to address risks and opportunities (6.1), which involves conducting AI risk assessment (6.1.2) and AI risk treatment (6.1.3). A key component is the AI System Impact Assessment (AISA) (6.1.4), which determines the potential consequences for individuals, groups, or societies.
- Operation (Clause 8): Focuses on implementing the planning, including AI risk management (8.2) and performing the AISA (8.4).
- Annex A (Normative Controls): Lists objectives and reference controls. Control A.9.3—Objectives for the responsible use of AI systems requires the organization to identify and document objectives to guide responsible use, including: fairness,

accountability, transparency, explainability, reliability, safety, robustness and redundancy, privacy and security, and accessibility. ISO/IEC 42001 defines "control" as a measure that maintains and/or modifies risk. Annex C of the standard details potential organizational objectives and AI-related risk sources, including Complexity of the environment, Lack of transparency and explainability, Level of automation, and Quality of data used in machine learning.

3.2 NIST AI RMF 1.0: Functions and Trustworthiness Characteristics

The NIST AI RMF 1.0 was developed as a voluntary, non-prescriptive resource to help organizations manage AI risks and promote responsible and trustworthy development and use. The NIST RMF is structured into four core functions that organize AI risk management activities [3]:

- GOVERN: A cross-cutting function that establishes the risk management culture, policies, responsibilities, and accountability structures. It includes managing legal and regulatory requirements (GOVERN 1.1) and integrating trustworthy AI characteristics into organizational policies (GOVERN 1.2).
- MAP: The function that establishes the context for framing AI risks. It involves understanding and documenting intended purposes, laws, standards, and expectations. It includes categorizing the AI system (MAP 2) and characterizing impacts (MAP 5).
- MEASURE: Applies tools, techniques, and methodologies to analyze, assess, and monitor AI risk and related impacts. It involves identifying and applying appropriate methods and metrics (MEASURE 1) and assessing AI systems for the trustworthiness characteristics (MEASURE 2).
- MANAGE: Involves allocating resources for mapped and measured risks, developing response and recovery plans, and communicating about incidents. A crucial subcategory is determining whether development or deployment should proceed (MANAGE 1.1) and prioritizing the treatment of documented risks.

NIST defines seven characteristics of trustworthy AI systems that, when addressed, can reduce negative risks: Valid and Reliable; Safe; Secure and Resilient; Accountable and Transparent; Explainable and Interpretable; Privacy-Enhanced; and Fair – with Harmful Bias Managed.

The NIST AI RMF recognizes that AI risks differ from traditional software risks due to data dependence, complexity, difficulty in predicting failure modes (emergent properties), and the difficulty in conducting rigorous testing, making the framework essential for managing these specificities.

3.3 Methodological Approach

The subsequent analysis employs a comparative mapping methodology (cross-referencing). The process was conducted in three sequential stages: (1) Deconstruction of CNJ Resolution 615/2025 into its fundamental normative requirements, including principles, governance mandates, and technical controls; (2) Exhaustive analysis of the ISO/IEC 42001 and NIST AI RMF 1.0 frameworks to identify corresponding clauses, controls, functions, and subcategories; and (3) Systematization of the mapping into tabular structures, justifying the correlation between each regulatory requirement and the international control mechanisms. This structured approach aims to ensure the traceability, replicability, and validity of the analysis, providing a clear roadmap for the operationalization of compliance.

4. Cross-Referencing Controls: 615/2025, ISO 42001, and NIST AI 100-1

The cross-referencing of CNJ Resolution 615/2025 with ISO 42001 and the NIST AI RMF 1.0 reveals a high degree of conceptual alignment, although the international standards offer the operational structure to implement and audit the Judiciary's regulatory requirements. Resolution 615/2025 itself endorses the use of these standards as a reference for security standards.

4.1 Governance and Accountability

CNJ 615/2025 Requirement	ISO/IEC 42001 Controls / Clauses	NIST AI RMF 1.0 Functions / Subcategories
Governance/Accountability: Definition of those responsible for oversight and accountability.	Leadership (Clause 5): Leadership and commitment; Roles, responsibilities, and authorities (5.3).	GOVERN 2: Accountability structures implemented; Documented and clear roles and responsibilities (GOVERN 2.1).
Documentation/Transparency: Publication of reports on functioning, data used, and supervision mechanisms.	A.4.2: Documentation of resources, including relevant information about AI activities and resources. A.8.5: Information for interested parties.	GOVERN 4.2: Teams document risks and potential impacts and communicate more broadly. MANAGE 4.3: Incidents and errors communicated to relevant actors.
Integration and Collaboration: Prioritize interoperable and collaborative solutions (Art. 12, IV).	A.3.3: Reporting of concerns (mechanism for reporting problems).	GOVERN 5: Processes for robust engagement with relevant actors.

Tab. 1: Governance requirement - CNJ 615/ISO 42001/NIST AI 1.0

CNJ requires that governance align with the law and normative acts is mirrored by NIST in GOVERN 1.1 (legal and regulatory requirements are understood, managed, and documented).

4.2 Risk Assessment and Mitigation

CNJ 615/2025 Requirement	ISO/IEC 42001 Controls / Clauses	NIST AI RMF 1.0 Functions / Subcategories
Preliminary Assessment and Risk Categorization: Define risk level (high/low) based on objective criteria and impact on fundamental rights.	Planning (Clause 6): Actions to address risks and opportunities (6.1). Context (4.1): Determine relevant external and internal values.	MAP 2: Categorization of the AI system is performed. MAP 5: Impacts on individuals, groups, and society are characterized (5.1: probability and magnitude).
Continuous Algorithmic Impact Assessment (AIA): Mandatory for high risk, identifying preventive and mitigating measures.	6.1.4: AI system impact assessment (AISA). 8.4: Perform AISA and retain documented information.	MAP 1.1: Understand potential positive and negative impacts of system uses. MEASURE 2: Assessment of trustworthiness characteristics.
Risk Treatment and Correction: Adoption of corrective measures for deviations.	6.1.3: AI risk treatment (determining necessary controls, see Annex A). 10.2: Nonconformity and corrective action.	MANAGE 1: Risks prioritized and managed. MANAGE 2: Strategies to minimize negative impacts (2.3: procedures for responding to unknown risk).
Human Supervision: Review and possibility of modification of any product generated by AI.	A.3.2: AI roles and responsibilities defined.	MANAGE 2.2: Mechanisms to suspend, deactivate, or disengage AI systems with inconsistent performance.

Tab. 2: Risk management - CNJ 615/ISO 42001/NIST AI 1.0

The mapping of risks in NIST (MAP) and the risk assessment in ISO (6.1.2) provide the formal methodologies to comply with Resolution 615/2025's requirement to perform a preliminary assessment and categorization.

4.3 Trustworthiness and Technical Attributes

CNJ 615/2025 Attribute	ISO/IEC 42001 (Annex C / A.9.3)	NIST AI RMF 1.0 (Trustworthiness Characteristics)
Justice, Equity, Non-Discrimination	C.2.5 Fairness (inappropriate application of automated systems can be unfair). A.9.3 Responsible use objectives include Fairness.	Fair – with Harmful Bias Managed. MEASURE 2.11 assesses fairness and bias.
Transparency and Explainability	C.2.11 Transparency and explainability (relates to how the system operates and explanations for outputs). A.9.3 Objectives include Transparency and Explainability.	Accountable and Transparent. Explainable and Interpretable.
Legal Certainty and Information Security	C.2.10 Security (in the context of AI and ML, new security concerns must be considered). ISO 27000 series are relevant.	Secure and Resilient. Aligned with the NIST Cybersecurity Framework.
Data Protection/Anonymization (Privacy by Design/Default)	C.2.7 Privacy (misuse of personal and sensitive data). A.7.2 Data for AI systems development and improvement (includes privacy).	Privacy-Enhanced. Use of PETs (Privacy-Enhancing Technologies) and data minimization methods.
Robustness and Reliability	C.2.8 Robustness (ability to maintain performance level under various circumstances). A.9.3 Objectives include Reliability and Robustness.	Valid and Reliable. Robustness is the ability to maintain performance under a variety of circumstances.

Tab. 3: Reliability attributes - CNJ 615/ISO 42001/NIST AI 1.0

Resolution 615/2025 details data treatment (Art. 27, 28, 30), requiring version control, secure storage, encryption, and anonymization at the source. These are specific controls that fit into the broader framework of ISO 42001, under A.7 Data for AI systems (which addresses data quality and provenance), and the Privacy-Enhanced practices of the NIST AI RMF. NIST, moreover, recognizes that AI presents new privacy risks by allowing inference to identify individuals.

5. Discussion and Implications for Responsible AI Engineering

The cross-referencing of CNJ 615/2025, ISO 42001, and NIST AI RMF 1.0 provides the theoretical and practical framework for the operationalization of the law. This mapping corroborate the

conception of Artificial Intelligence as a sociotechnical system, wherein outcomes and risks emerge from the complex interplay between algorithms, data, human actors, and institutional contexts. The CNJ Resolution establishes the societal and ethical objectives, while the ISO and NIST frameworks provide the management and control mechanisms. The analysis demonstrates that responsible AI governance cannot be achieved through abstract ethical principles alone; it demands a formal management infrastructure and auditable engineering processes, as provided by international standards.

5.1 Operationalization

Resolution 615/2025 requires AI solutions to be audited from various angles and that the verification of compatibility with fundamental rights occur at all phases of the life cycle. The NIST AI RMF 1.0 positions Testing, Evaluation, Verification, and Validation (TEVV) as tasks performed throughout the entire AI life cycle. The NIST MEASURE function details how these activities should be carried out, ensuring that test sets are documented (MEASURE 2.1) and that the system is demonstrated to be valid and reliable (MEASURE 2.5).

For the Judiciary, the adoption of TEVV processes, as detailed in NIST, allows the courts to demonstrate, in an objective, repeatable, and scalable manner, that AI solutions, classified as high or low risk, meet the standards of reliability, robustness, and non-discrimination required by Resolution 615/2025.

5.2 Management of Harmful Biases

The CNJ is emphatic about the prevention of illegal or abusive discriminatory biases (Art. 8º). The NIST AI RMF 1.0 provides the context for managing bias, recognizing three main categories: systemic, computational/statistical, and human-cognitive. The NIST framework, by emphasizing that AI development should be conducted by diverse teams (demographically and disciplinarily) (GOVERN 3.1), offers a direct governance mechanism to combat systemic and human-cognitive bias, aligning with the CNJ's directive on the pursuit of diversity and representativeness in the composition of development teams (Art. 35).

5.3 ISO 42001 in Regulatory Compliance

While Resolution 615/2025 and the NIST RMF focus on AI-specific risk assessment and mitigation and the promotion of trust, ISO/IEC 42001 provides the management system framework (AIMS) necessary for continuous compliance and organizational auditability. The AIMS ensures that the organization (the court or vendor) incorporates regulatory requirements (like those of the CNJ) into its policies (5.2), plans (6.1), and operations (8.1). ISO 42001 is designed to help organizations meet objectives and obligations related to the use of AI systems. Certification in an AIMS can serve as formal evidence that the court or vendor complies with the requirements of Resolution 615/2025, especially those related to documentation, risk management, competence, and continual improvement (Clauses 7, 9, 10).

6. Conclusion

The CNJ's technological resolutions establish rigorous controls for the use of AI in the Brazilian Judiciary, where risks and benefits emerge from the interaction of technical aspects and social factors, demanding a robust management framework [4-9]. ISO/IEC 42001:2023 and the NIST AI RMF 1.0 provide this framework. The cross-referencing demonstrated that:

- The regulatory mandates of the CNJ, such as the need for Algorithmic Impact Assessment and risk categorization, find practical and procedural implementation in the MAP and MEASURE functions of NIST and the Planning and Operation activities of ISO 42001.
- The ethical principles required by the Resolution, such as fairness, explainability, and security, are directly mapped to the Trustworthiness Characteristics of NIST and the Responsible Use Objectives of ISO 42001 (A.9.3).
- ISO/IEC 42001 offers the formal management system that allows courts to incorporate and audit compliance with Resolution 615/2025, ensuring that AI risk management is integrated.

In summary, the adoption of the controls and methodology offered by ISO 42001 and the NIST AI RMF 1.0 enables the Brazilian Judiciary not only to comply with the directives of Resolution 615/2025 but also to achieve a level of Responsible AI Engineering (RAIE) that increases the trustworthiness of AI systems throughout their entire life cycle.

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